



Rhea Soil Nutrient Prediction Challenge

04 February - 09 March 2026

Competition Debrief | 27 March 2026



Challenge Summary

Rhea supports smallholder farmers by providing accessible, real-time soil testing and data driven insights. Improved soil information enables:

- Targeted fertiliser application for higher yields
- Reduced input costs through efficient nutrient use
- Improved resilience through better soil management
- More sustainable and profitable farming

The challenge objective was to predict key soil nutrient levels (Al, Bl, ... etc) using geospatial and soil data.

PRIZE POOL | € 8 250

Rhea Soil Nutrient Prediction Challenge

Can you predict soil nutrient
content when lab tests
aren't an option?

DATA
GOVERNANCE
IN AFRICA

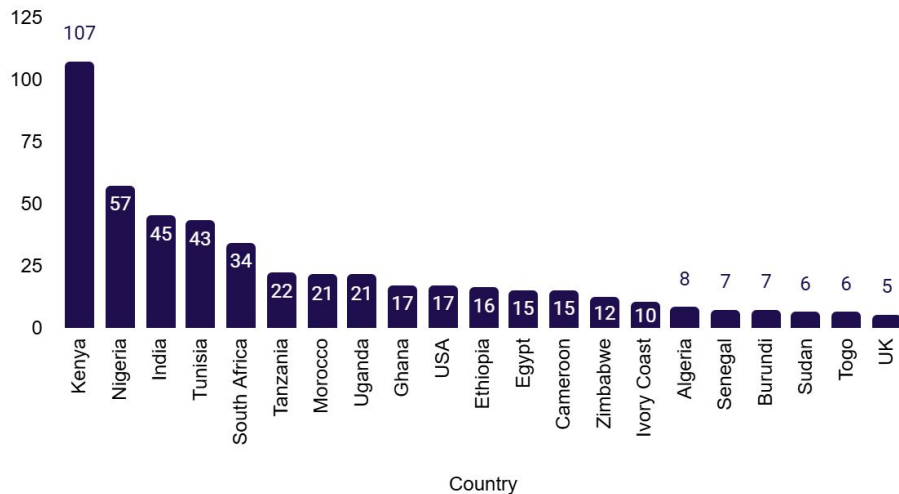
Challenge Statistics

- › **610** users engaged
- › **18%** female
- › **82** countries (**38** African countries)
- › **4 441** submissions
- › **15 000+** estimated hours of data science work supported



Country Representation

Top 20 participation by country



Country distribution showed strong engagement from a diverse set of African countries, which was a positive outcome for this challenge.

- Country participation is often higher when a challenge is specifically targeted at certain countries or regions - Kenya in this case.
- Good participation from Francophone countries such as, Cameroon, Côte d'Ivoire likely due to Digital Africa geographic focus.

Winners



Yisak Birhanu
[Yisakberhanu](#)

Top overall model

1st overall / 2nd top Zindi performer

Country: Ethiopia

Work: ML Engineer

Medals: 17th Gold

Score: 0.5556



L. Gladys & P. Kamau
[BitsnBytes](#)

Top-placed female team

Country: Kenya

Work: Data Scientist & Research
Engineers

Medals: 3rd & 7th Gold

Score: 0.4839



Jackson Kahungu
[keystats](#)

Top Kenyan

Country: Kenya

Work: Data Scientist

Medals: 7th Gold

Score: 0.4727

[Your winning solutions](#)

1ST PLACE SOLUTION OVERVIEW (TOP OVERALL)

Core Strategy: Domain-Aware Ensemble

Problem: Standard models fail to generalize due to a strong temporal shift between training and test data.

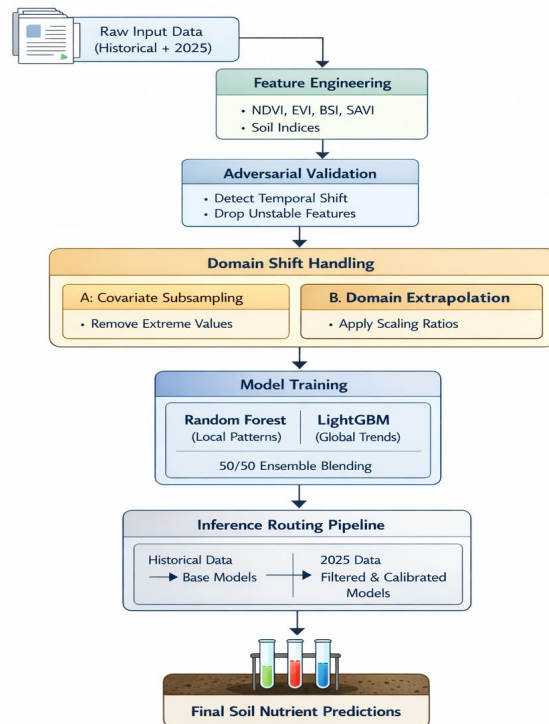
Solution & Methodology

- Adversarial feature selection → retain stable features only
- Domain shift correction via subsampling and extrapolation scaling
- Model Design: Random Forest + LightGBM in a 50/50 ensemble

Inference & Insights:

- Inference: separate pipeline for historical and 2025 data
- Key Insight: Treat domain shift as a model problem, not post-processing

Domain-Aware Ensemble for Soil Nutrient Prediction



2ND PLACE SOLUTION OVERVIEW (TOP FEMALE)

Core Strategy: Multi-Source Spatial Ensemble

Problem

- Predict 13 soil nutrients across Africa with sparse ground data

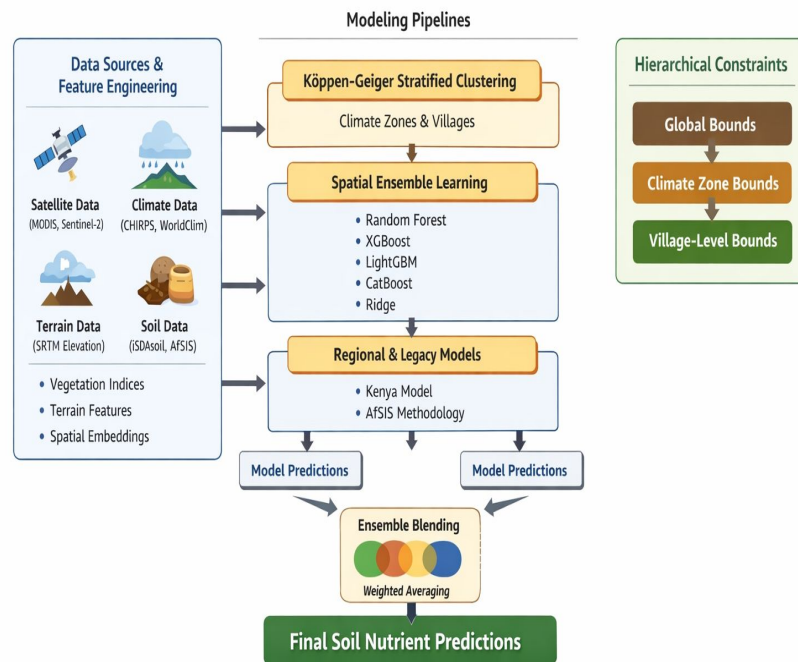
Solution

- Satellite, climate, and soil datasets combined
- Spatially-aware features (indices, terrain, neighbourhoods)

Modelling

- Climate-based clustering (Köppen–Geiger zones)
- Village-level modelling with hierarchical statistics
- Multi-model ensemble: Random Forest, XGBoost, LightGBM, CatBoost, Ridge

Key Insight: Spatial structure + ensemble diversity = strong generalisation.



3RD PLACE SOLUTION OVERVIEW (TOP KENYAN)

Core Strategy: Multi-Scale Geospatial Ensemble

Problem

- Predict 13 soil nutrients across Africa without lab data

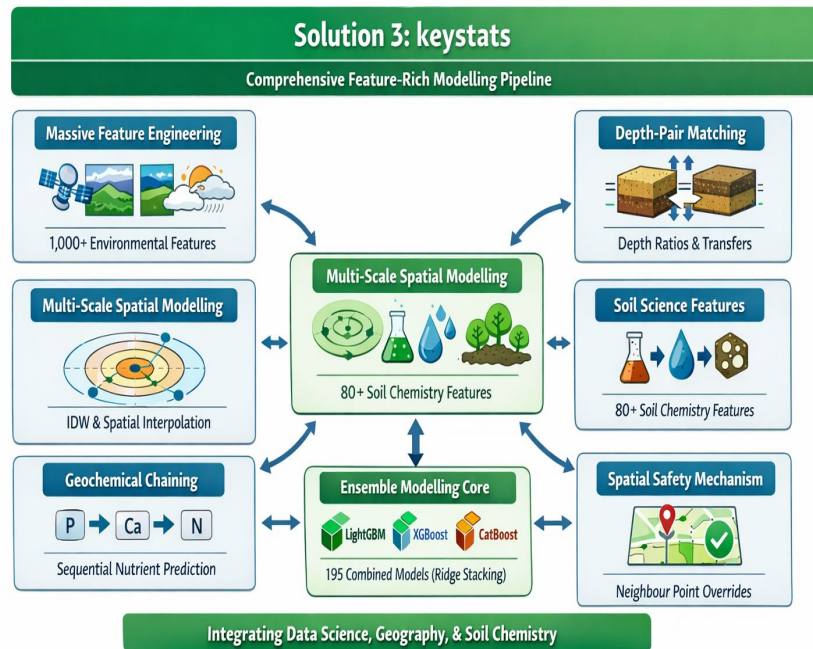
Solution

- 1,000+ features from satellite, climate, and soil data
- Depth-pair matching (64% shared locations)
- Multi-scale spatial modelling
- Domain knowledge integration (Soil Chemistry)

Modelling

- Per-nutrient pipelines; soil science-informed features
- LightGBM, XGBoost, CatBoost (195 models); Ridge stacking

Key Insight: Spatial structure + domain knowledge + ensemble learning drives strong generalisation.





Lessons learnt

- **Data acquisition and preparation process** was smooth, supported by Rhea provision of well-structured tables with clearly defined unique keys, which simplified integration, cleaning, and feature engineering.
- **Diversity of approaches:**
 - Time-aware modelling vs spatial modelling vs domain-driven + data exploitation
 - Lean ensemble vs multi-pipeline ensemble vs large-scale optimised ensemble
- **Challenge Complexity & Evaluation**
 - Compute-heavy training
 - Multi-metric scoring
 - Weighted 13 RMSEs (per nutrient)
 - Weights $\propto$ target availability

Feedback on Solutions

- Were any solutions unclear?
- What were you impressed with?
- Where were you disappointed?
- How can we address this?





What's next?

